

Selenium with Java Course Syllabus

Archer Infotech, Kothrud Pune | Classroom & live-online batches | Updated September 2026

Institute: Archer Infotech, Kothrud, Pune - IT training since 2009

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Mode: Classroom at Kothrud, Pune and live-online batches

Duration: 2 Months

Level: Intermediate

Course page: <https://archerinfotech.in/courses/testing-qa/selenium-training-in-pune>

Course Overview

Selenium with Java is Pune's most-asked-for automation testing track in 2026, sitting at the intersection of QA careers and software engineering. This 2-month programme takes you from Java fundamentals through Selenium 4 WebDriver, TestNG, the Page Object Model, Maven dependency management, and Jenkins CI integration — finishing with a deployed automation framework you can show in interviews. Designed for manual QA testers moving to automation and for engineering graduates targeting Pune's services-sector QA pipeline.

Why Learn Selenium with Java in Pune in 2026

Pune's IT services sector — Persistent, Capgemini, Mindtree, Tech Mahindra, Cognizant, Wipro — runs the majority of its automation testing on Selenium with Java. Walk through any of their Pune campuses and the QA tooling stack is the same: Selenium 4 + TestNG, Maven build, Jenkins pipeline, Page Object Model framework, Cucumber BDD on top for stakeholder-readable scenarios. This isn't a trend; it's the established enterprise default and it isn't changing soon. Hiring volume for Selenium engineers in Pune ran consistently in the 600–900 active job listings range every month of 2025 across Naukri, LinkedIn, and Indeed.

The career economics work too. Pune Selenium fresher salaries currently sit Rs. 3.5–4.5 LPA at services majors and Rs. 4.5–6 LPA at product companies (sampled from AmbitionBox + Indeed Pune QA Automation listings, last 12 months). One to two years of experience plus a clean automation portfolio moves you to Rs. 6–10 LPA. Senior SDET and Test Architect tracks in Pune product companies pay Rs. 14–22 LPA. Because the role sits inside engineering teams now — not in a separate QA org — career mobility into developer roles is genuinely available; we've seen multiple Archer Infotech automation alumni move into backend or full-stack roles within two years.

Selenium 4 brought meaningful improvements over Selenium 3: native W3C protocol support, relative locators (above, below, near), built-in observability hooks, and proper TypeScript / async APIs. Older Selenium 2/3 training that you'll find on free YouTube tutorials skips these — leaving graduates with framework knowledge that doesn't match what hiring managers expect when they screen for Selenium in 2026.

- 600–900 active Pune Selenium job listings each month (2025)
- Pune fresher band Rs. 3.5–4.5 LPA services / Rs. 4.5–6 LPA product
- 1–2 years + portfolio = Rs. 6–10 LPA range
- Senior SDET / Test Architect in Pune product cos = Rs. 14–22 LPA
- Selenium 4 (current) — relative locators, W3C protocol, BiDi APIs
- Pairs naturally with REST Assured (API automation) + Cucumber (BDD)

Who This Course Is For

This course fits you if

- Manual QA testers (1+ years) who want to break into automation before being squeezed out of the role
- Engineering / BCA / BSc-CS graduates targeting Pune services-sector QA roles where Selenium is the default ask
- Java developers wanting to pivot into SDET / Test Architect tracks (the highest-paid QA segment)
- Bootcamp graduates from manual testing programmes who need an automation specialisation
- Working professionals (weekend batch) who can commit ~6 hrs/week for 2 months

This course is not the right fit if

- Complete coding beginners with zero Java exposure — start with our Java track first, then Selenium next
- Anyone targeting product-company SDET roles at FAANG-tier compensation — those need DSA + system design beyond what an automation course covers
- QA managers looking for a strategic overview without writing code — this course is hands-on framework building, not theory
- People who want to bypass interview prep — Selenium framework knowledge alone doesn't close offers; you'll do mock interviews + frameworks both

Detailed Curriculum

Module 1: Software Testing Fundamentals (Week 1)

The grounding every automation engineer is assumed to have and is quietly judged on in interviews. What testing is for, SDLC and STLC as processes rather than acronyms, and the verification-versus-validation distinction that panels still ask about.

Then the daily working vocabulary: functional and non-functional testing, smoke, sanity and regression testing and when each runs in a release cycle, writing test scenarios and test cases someone else can execute, the defect life cycle, and severity versus priority — which candidates routinely conflate. Agile, Scrum and Jira close the module, because that is the process every Pune QA role operates inside.

- Introduction to software testing
- SDLC and STLC
- Verification and validation
- Functional and non-functional testing
- Smoke, sanity and regression testing
- Test scenarios and test cases
- Defect life cycle
- Severity and priority
- Agile and Scrum fundamentals
- Jira for test management

Module 2: Core Java for Test Automation (Weeks 1–2)

Java scoped to what test automation actually uses, taught from the beginning. Fundamentals, variables and data types, operators, conditionals and loops, then arrays and strings — the structures test data lives in — and methods.

Object-oriented programming is the centre of the module, because the Page Object Model is unusable without it: classes and objects, constructors, inheritance, polymorphism, abstraction and interfaces. Exception handling, the Collections framework and file handling close it, covering test-data reading and failing informatively rather than with a raw stack trace.

- Java fundamentals and the JDK setup
- Variables and data types
- Operators
- Conditional statements
- Loops
- Arrays and strings
- Methods
- Classes and objects
- Constructors
- Inheritance

- Polymorphism
- Abstraction
- Interfaces
- Exception handling
- Collections framework
- File handling

Module 3: Selenium WebDriver (Weeks 3–4)

The core automation skill. Selenium's architecture and the W3C WebDriver protocol, driver setup and browser automation, then locators in depth — id, name, class, link text, CSS selectors and XPath — with the discipline of choosing locators that survive a UI change rather than ones that merely work today.

The module then works through everything a real page throws at you: forms, dropdowns, checkboxes and radio buttons, alerts, frames, multiple windows and tabs, web tables, and mouse and keyboard actions. Synchronisation gets particular attention — implicit, explicit and fluent waits, and why a fixed sleep is the first thing a reviewer flags. JavaScript executor, screenshots and file upload and download close it.

- Selenium architecture and the W3C protocol
- WebDriver setup and driver management
- Browser automation
- Web elements and element interactions
- Locators — id, name, class, link text
- XPath and CSS selectors
- WebDriver commands
- Handling forms
- Dropdowns
- Checkboxes and radio buttons
- Alerts
- Frames
- Multiple windows and tabs
- Web tables
- Mouse and keyboard actions
- Synchronization and waits
- JavaScript executor
- Screenshots
- File upload and download

Module 4: TestNG (Week 4)

The test runner that turns a set of scripts into an orchestrated suite. TestNG's architecture and annotation lifecycle, assertions — hard versus soft, and when each is the right call — and test groups for selective execution.

Parameterisation and DataProvider drive data-driven runs; test dependencies express ordering where it genuinely exists rather than by accident. Parallel execution and the TestNG XML suite file close the module, including thread-safety of the driver instance, which is the defect that appears the moment a suite first runs in parallel.

- TestNG architecture
- TestNG annotations and lifecycle
- Assertions — hard and soft
- Test groups
- Parameterization
- DataProvider for data-driven tests
- Test dependencies
- Parallel execution and driver thread safety
- TestNG XML
- Test suites

Module 5: Maven and Dependency Management (Week 5)

The build tool every Java automation project is delivered through, and a frequent gap in self-taught profiles. Maven fundamentals and the convention-over-configuration project structure, then `pom.xml` — dependencies, versions, scopes and the transitive resolution that explains mysterious version conflicts.

Plugins cover Surefire for test execution and the compiler plugin for source levels. The build lifecycle — validate, compile, test, package, install — is taught as the sequence CI will run, and the module ends by running the suite entirely from the command line, which is the prerequisite for the Jenkins module.

- Maven fundamentals
- Project structure and conventions
- `pom.xml` anatomy
- Dependencies, versions and scopes
- Transitive dependencies and conflict resolution
- Plugins — Surefire and compiler
- Build lifecycle
- Running automated tests from the command line

Module 6: Automation Framework Development (Weeks 5–6)

The module that separates someone who writes tests from someone who owns a suite. The Page Object Model implemented properly — page classes, locators as fields, methods expressing user intent rather than clicks — plus Page Factory and its initialisation model.

Then the framework patterns and an honest account of each: data-driven, keyword-driven and hybrid, including when keyword-driven is genuinely worth its indirection and when it is over-engineering. Configuration management, utility classes, reusable components, logging, exception handling and screenshot management complete an architecture a new joiner can navigate.

- Page Object Model
- Page Factory
- Data-driven framework
- Keyword-driven framework
- Hybrid framework
- Configuration management
- Utility classes
- Reusable components
- Logging
- Exception handling
- Screenshot management

Module 7: BDD with Cucumber (Week 6)

Behaviour-driven development as it is actually practised, including a frank account of when it earns its overhead. BDD fundamentals and the three-amigos idea, then Gherkin — Given, When, Then — and feature files written to be readable by someone who is not a programmer.

Scenarios and scenario outlines for data variation, step definitions mapping plain language to Java, hooks for setup and teardown, and tags for selective execution. The module integrates Cucumber with the Page Object framework built previously, so BDD becomes a layer on top rather than a second parallel suite, and closes on Cucumber reporting.

- Introduction to BDD and the three-amigos practice
- Gherkin syntax
- Feature files
- Scenarios
- Scenario outlines
- Step definitions and step reuse
- Hooks

- Tags and selective execution
- Cucumber with Selenium and the Page Object layer
- Cucumber reports

Module 8: Reporting (Week 7)

Reports are how a QA engineer communicates with people who will never read the code, so they are treated as a deliverable rather than a switch. The default TestNG reports for a quick local view, then Extent Reports and Allure — the two formats Pune teams most often standardise on — with suites, steps, attachments and history.

Screenshots attached on failure turn a red test into an actionable one. The module ends on what a test-execution report should actually say to a delivery manager versus a developer, which is a communication skill rather than a tooling one.

- TestNG reports
- Extent Reports
- Allure Reports
- Screenshots on failure
- Attaching logs and environment metadata
- Test execution reports for different audiences

Module 9: Git and GitHub (Week 7)

Version control as the automation engineer uses it. Git fundamentals — working tree, staging and commits — then the day-to-day commands: status, diff, log, add, commit, push and pull.

Branching and merging with conflict resolution done without losing work, then the GitHub layer: repositories, pull requests, and code review applied to test code, which is code and deserves the same treatment. The module closes on maintaining an automation project as a shared asset — .gitignore for drivers, target folders and reports, commit hygiene, and a README that lets a reviewer actually run your suite.

- Git fundamentals
- Repository management
- Commit, push and pull
- Branching and merging
- Conflict resolution without losing work
- GitHub collaboration and pull requests
- Code review applied to test code
- Maintaining automation projects on GitHub

Module 10: CI/CD with Jenkins (Week 8)

Getting the suite running without you. CI/CD fundamentals and why a suite that only runs on the author's laptop delivers a fraction of its value. Jenkins setup, job creation, and running Selenium tests from Jenkins with the right JDK and browser configuration on the agent.

Maven integration invokes the build lifecycle from the previous module; GitHub integration triggers runs on push and pull request; scheduled execution covers nightly regression. Pipeline fundamentals close the module, along with the operational reality of a suite that fails at 2am — notifications, retries and triage.

- CI/CD fundamentals for test automation
- Jenkins setup
- Creating Jenkins jobs
- Running Selenium tests from Jenkins
- Maven integration
- GitHub integration and build triggers
- Scheduled test execution
- Pipeline fundamentals
- Notifications, retries and failure triage

Module 11: AI-Assisted Testing (Week 8 + capstone)

The capability Pune QA panels now probe for directly, taught with the validation discipline that makes it professional rather than reckless. Generative AI applied across the testing workflow: producing test scenarios, generating test cases, and converting existing manual test cases into runnable Selenium automation.

Then the code-level uses — generating Selenium scripts, XPath and locator suggestions, page objects and framework scaffolding — followed by the analytical ones: AI-assisted debugging, test-data generation, defect analysis and summarisation, log analysis, and refactoring automation code. The module is explicit throughout that AI output is a draft: it closes on prompt engineering for test engineers, using coding assistants inside an automation workflow, and the best practices and limitations that decide whether AI-generated tests help a suite or quietly rot it.

- Introduction to generative AI for testing
- Using AI to generate test scenarios
- AI-assisted test case generation
- Generating Selenium scripts with AI
- AI-assisted XPath and locator generation
- Converting manual test cases into automation scripts
- AI-assisted debugging
- Test data generation using AI
- Defect analysis and summarization
- Log analysis using AI
- Generating page objects with AI
- Refactoring automation code with AI
- AI-assisted framework development
- Prompt engineering for test engineers
- Using coding assistants in automation testing
- Best practices and limitations of AI-generated tests

Capstone Projects

E-commerce Checkout Flow Automation

Build a complete end-to-end Selenium framework that automates the critical checkout flow of a public e-commerce demo site — login, search, add to cart, address selection, payment-step validation, order confirmation. Page Object Model architecture, data-driven for 5+ product categories, Jenkins-runable, ExtentReports HTML output. The portfolio piece that shows up best on a fresher CV.

- Technology: Selenium 4 WebDriver
- Technology: Java
- Technology: TestNG
- Technology: Maven
- Technology: PageFactory
- Technology: Apache POI (Excel data)
- Technology: ExtentReports
- Technology: Jenkins

Banking Application Login + Transaction Flow

A more complex framework with multi-window handling (banking apps often open transactions in new tabs), alert handling, and CAPTCHA-bypass strategies for test environments. Covers parameterised tests across 3 user roles (customer, teller, admin) with role-specific permissions verification. Includes a Slack notifier on build failure — useful narrative point at interviews.

- Technology: Selenium 4
- Technology: Java
- Technology: TestNG
- Technology: Cucumber BDD
- Technology: Log4j2

- Technology: Properties config
- Technology: Slack webhook integration
- Technology: Jenkins pipeline

Capstone — Cross-Browser CI Framework with API Layer

Full framework combining Selenium UI tests (cross-browser via Selenium Grid, parallel execution) AND REST Assured API tests (auth-token retrieval, then UI tests that use the same authenticated session). Pushed to GitHub with a proper README, runs via Jenkins on every merge to main. This is the project that demonstrates SDET-level breadth, not just Selenium-button-clicker breadth.

- Technology: Selenium 4
- Technology: Selenium Grid
- Technology: Java 17
- Technology: TestNG
- Technology: Cucumber
- Technology: REST Assured
- Technology: Apache POI
- Technology: Jenkins (Maven pipeline)
- Technology: GitHub
- Technology: Maven

Career Outcomes

Pune's QA Automation market has bifurcated into two clear bands. The services-sector pipeline (Persistent, Capgemini, Mindtree, Tech Mahindra, Cognizant, Accenture, IBM India) hires Selenium engineers at scale — usually as part of a 3-6 month bench-to-project ramp. Salaries here start at Rs. 3.5–4.5 LPA fresher; an ISTQB Foundation cert plus a Selenium portfolio bumps the offer Rs. 0.5–1 LPA. Product-company SDET roles (Persistent product teams, Cybage, BrowserStack, GUVI, Helpshift, Avaamo, Druva) pay 30–60% more for the same headline experience because the role expectations include CI/CD ownership, API testing, and occasional dev tasks — exactly what this course's Module 5 + Module 6 prepare you for.

The hiring funnel at Archer Infotech has a consistent pattern for Selenium graduates: ~70% receive interview calls within 30 days of course completion, ~50% have an offer in hand within 60 days. The bottleneck is almost never Selenium framework knowledge (the course handles that); it's interview communication and DSA basics for product-company contexts (where the placement prep modules earn their keep). We don't claim 100% placement — our institute-records placement rate is 90% across all tracks and Selenium runs close to that average.

Roles after this course

- QA Automation Engineer
- Selenium Test Engineer
- SDET (Software Development Engineer in Test)
- Test Automation Lead
- QA Automation Architect (with experience)
- DevOps + QA Hybrid (with CI/CD specialisation)

Indicative salary bands

- QA Automation Engineer (fresher) - Rs. 3.5–5 LPA (services) / Rs. 4.5–6 LPA (product) (source: AmbitionBox Pune QA Automation (last 12 mo))
- QA Automation Engineer (1–3 yrs) - Rs. 6–10 LPA (source: Indeed Pune QA Automation listings (last 12 mo))
- SDET (3–6 yrs) - Rs. 10–18 LPA (source: Glassdoor Pune SDET)
- Test Architect / Sr SDET (6+ yrs) - Rs. 14–22 LPA (source: AmbitionBox Pune Test Architect)

Companies hiring for these skills

- Persistent Systems
- Capgemini

- Mindtree (LTI Mindtree)
- Tech Mahindra
- Cognizant
- Wipro
- Accenture
- Cybage
- BrowserStack
- Helpshift
- Druva
- GUVI
- Avaamo
- Saksoft
- TCS
- IBM India

Batch Modes and Duration

Duration: 2 months (8 weeks) for the standard weekday/online track; ~10 weeks for the weekend track

Classroom

- Location: Archer Infotech Kothrud campus (Flat No. 12, Divyadarshan Housing Society, Kothrud, Pune 411038)
- Morning batch: Monday–Friday 10:00–11:30
- Evening batch: Monday–Friday 19:00–20:30
- Saturday lab session: 10:00–13:00 (3-hour deep practice)

Live online

- Live sessions: Monday–Friday 19:30–21:00 IST
- Recorded sessions available in LMS within 24 hrs of live session
- Google Meet for live sessions
- GitHub for code distribution
- Slack channel for batch-wide doubts

Weekend

- Saturday + Sunday 10:00–13:00 (3 hrs/day, 6 hrs/week)
- Weekend track runs 10 weeks instead of 8 to maintain total contact hours

Batch sizes are kept under 18 for the weekday/online tracks and under 12 for the weekend track — small enough that you'll know the trainer by name and they'll know your project repo. New batches typically start every 3–4 weeks; check the live batch schedule.

Fees

Selenium with Java track is priced in the mid-band of our catalogue — reflective of the 2-month duration and the per-batch lab + Jenkins server infrastructure. EMI plans available; contact admissions for the current fee structure.

Indicative range: Rs. 25,000 – Rs. 35,000 (typical track band)

- One-time payment (5% discount)
- EMI: 50% at enrolment + 50% at 4 weeks
- EMI: 3-month plan (1/3 monthly)

Placement Support

Placement support is bundled — no separate fee. Selenium track graduates plug into the same Archer placement cell that supports our Java, Python, and Full Stack tracks, which has 17 years of relationships with Pune QA hiring managers. The placement workflow starts in Week 6 (parallel to your capstone build), not after the course finishes — by the time you've

graduated, your CV is reviewed, your GitHub profile is portfolio-ready, and you've done at least 2 mock interviews.

We don't guarantee placement. Our institute-records placement rate is 90%, the Selenium track runs in that range, and the variance is mostly explained by interview communication readiness rather than framework knowledge. The placement cell expects you to apply to 15+ companies actively in the 60-day post-completion window; we open the doors and prepare you, but you walk through.

The placement process

- Week 6: CV review + GitHub portfolio audit + resume formatting for Pune services-sector and product-company tracks
- Week 7: First mock interview (technical) + interviewer feedback session
- Week 8: Second mock interview (HR + communication) + offer-negotiation primer
- Post-completion Week 1–2: Direct introductions to 10+ partner companies actively hiring Selenium
- Post-completion Week 3–8: Weekly placement-cell check-ins + interview-prep coaching as you apply

Partner companies

- Persistent Systems
- Capgemini
- MindTree (LTIMindtree)
- Tech Mahindra
- Amdocs
- Cybage
- Atos Syntel
- Saksoft
- Druva
- 100+ partner companies across Pune IT services + product sector

Prerequisites and How to Start

The course assumes you can read basic code (loops, if-statements, function calls) and have used a command line at least occasionally. We don't assume any Java syntax knowledge, any Selenium knowledge, or any framework-building experience. About 40% of each batch comes from manual QA backgrounds with 1+ years experience; ~35% are fresh BE / BCA / BSc-CS graduates; the remaining ~25% are Java developers pivoting to SDET tracks. The pace adjusts naturally — manual QA folks tend to need more support on the Java OOP modules but accelerate through framework design; fresh graduates need more hand-holding on framework patterns but pick up Java fast.

Before you enrol, do the 5-step starting sequence below. None of it is gated — if you've done these, you'll be productive in the first session.

Suggested first steps

- Install Java 17 + Maven on your machine — verify with `java -version` and `mvn -version`
- Install IntelliJ IDEA Community Edition (the IDE we use throughout the course)
- Create a free GitHub account if you don't have one and learn the basic clone / commit / push workflow
- Read the Selenium 4 introduction page on selenium.dev (~15 minutes) to know what you're about to learn
- Skim Chapter 1 of any free Java tutorial (Codecademy / Java Brains / W3Schools) — just enough to recognise classes and methods

Frequently Asked Questions

Do I need any prior automation testing or Java experience?

No. The course starts with a focused 'Java for Testers' module covering exactly what you need for Selenium, then layers Selenium fundamentals on top. About 35% of each batch are fresh graduates with zero Java background; they finish at the same place as the manual-QA cohort. The honest prereq is willingness to write code daily and complete the lab assignments — not pre-existing knowledge.

What's the actual difference between Selenium 3 and Selenium 4 — do interviewers ask about this?

Yes, they ask. Selenium 4 (current) uses the W3C protocol natively, adds relative locators (above/below/near), brings WebDriver BiDi APIs for real-time observability, and improved the wait API. Many YouTube tutorials still teach Selenium 3 — leaving graduates unable to answer 'what changed in Selenium 4' which is a screening question at most Pune services majors in 2026. This course is Selenium 4 throughout.

Will I be ready for SDET (Software Development Engineer in Test) roles after this course?

Ready to apply, yes. Closing offers at the SDET band typically also requires comfort with CI/CD pipeline ownership, REST API automation, and basic DSA — the course covers the first two in modules 5 and 6. DSA is on you; we point graduates to a focused 30-day NeetCode + system-design-primer track for the SDET interview prep. Realistic timeline: course graduation -> QA Automation Engineer offer in 1–2 months -> SDET pivot after 12–18 months in role.

How does the placement process work and how soon do you start applying me?

Placement-cell work starts Week 6 (parallel to your capstone) — CV review, GitHub polish, mock interviews. Direct introductions to 10+ partner companies start the week after course completion. We don't guarantee placement; our institute-records placement rate is 90% across all tracks. The Selenium track runs in that range. Variance is mostly explained by interview communication and the candidate's own application volume; we open the doors but you apply to 15+ companies actively in the 60-day post-completion window.

What tools and software do I need on my own laptop?

Java 17 (free), Maven (free), IntelliJ IDEA Community Edition (free), and Git (free). Optional but useful: Postman for REST API practice (free), and a GitHub account for portfolio hosting (free). No paid licences anywhere in the toolchain. Laptop spec — 8GB RAM minimum, 16GB recommended for running Selenium Grid + multiple browsers in parallel. SSD strongly recommended.

Can I move from manual QA to automation after this course?

That's exactly the most common transition path our Selenium graduates take. Manual QA testers typically need 2–3 months of post-course on-the-job practice before they feel fully confident in automation reviews, but offers come in within the same timeline as fresh-grad placements. The advantage manual testers have: they know the testing domain (test design, defect lifecycle, business workflows) so they catch test logic issues that pure developers miss. Combine that with Selenium framework skill and you're hireable above the 'pure-automation-engineer' band.

Why Selenium with Java instead of Selenium with Python?

Pune's services-sector enterprise default is Selenium with Java; that's where the 600–900 monthly listings sit. Selenium with Python has its niche (mostly product startups, data engineering test pipelines) but the Pune hiring volume is roughly 5x smaller. If your goal is maximum Pune job-market access, Java is the right pick. If your goal is parallel-skilling alongside Python development, ask the admissions team — we have a Python+Selenium variant we can route to.

What about Cucumber, TestNG, JUnit — which is taught and why?

TestNG is the primary test runner (Pune services-sector default). Cucumber BDD is covered in Module 6 because Pune BFSI and Insurance clients require Gherkin-readable scenarios. JUnit gets a comparison mention but isn't the primary runner — we focus depth on what hiring managers expect, which is TestNG. All three concepts (annotations, lifecycle, parameterisation) transfer between runners; switching is days, not weeks.

Enquire or book a free demo session. Archer Infotech, Flat No. 12, Divyadarshan Housing Society, behind Kothrud Bus Stand Road, Kothrud, Pune 411038. Phone +91 9850 678451. Email info@archerinfotech.in. Web archerinfotech.in.